

The effects of grapefruit on weight and insulin resistance: relationship to the metabolic syndrome.

To study the effects of grapefruit and grapefruit products on body weight and metabolic syndrome, 91 obese patients were randomized to either placebo capsules and 7 ounces (207 mL) of apple juice, grapefruit capsules with 7 ounces (207 mL) of apple juice, 8 ounces (237 mL) of grapefruit juice with placebo capsule, or half of a fresh grapefruit with a placebo capsule three times a day before each meal. Metabolic syndrome parameters were measured at the beginning and end of 12 weeks. After 12 weeks, the fresh grapefruit group had lost 1.6 kg, the grapefruit juice group had lost 1.5 kg, the grapefruit capsule group had lost 1.1 kg, and the placebo group had lost 0.3 kg. The fresh grapefruit group lost significantly more weight than the placebo group ($P < .05$). A secondary analysis of those with the metabolic syndrome in the four treatment groups demonstrated a significantly greater weight loss in the grapefruit, grapefruit capsule, and grapefruit juice groups compared with placebo ($P < .02$). There was also a significant reduction in 2-hour post-glucose insulin level in the grapefruit group compared with placebo. Half of a fresh grapefruit eaten before meals was associated with significant weight loss. In metabolic syndrome patients the effect was also seen with grapefruit products. Insulin resistance was improved with fresh grapefruit. Although the mechanism of this weight loss is unknown it would appear reasonable to include grapefruit in a weight reduction diet.